



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India



R V College of Engineering

Computer Science and Engineering

Field Visit's to Kolar, Karnataka

Real-Time Bat Acoustic Monitoring with ML Classification

In collaboration with Bat Conservation India Trust (BCIT) and Centre for Wildlife Studies (CWS)

Faculty Advisor:

Dr. Anala M R, Professor (ISE Dept.)

Student Coordinator:

Nihal Soni (1RV23CS152)

Nikhil Parashuram Bakale (1RV23CS152)



Brief Overview

As part of a socially responsible field-based learning initiative, we conducted multiple visits to **Kolar district, Karnataka**, including **Devarayanasamudra Hill** and **Hanumanalli Hill**.

The visits were organized in collaboration with **Bat Conservation India Trust**, a non-governmental organization working towards the protection of bat species and their natural habitats.

These regions are known to host **endangered and ecologically significant bat species**, whose populations are increasingly threatened due to habitat loss, human interference, and lack of awareness. The primary objective of this field activity was to **support conservation efforts through technology while developing social and environmental responsibility among students**.

Purpose and Objectives

Purpose

- Bats play an important role in the environment by helping in **pollination, seed dispersal, and insect control**.
- Many bat species in the Kolar region are endangered and poorly monitored.
- Bat Conservation India Trust (BCIT) required a device that can capture bat audio and make the data accessible online anytime through a user interface, even from remote locations.
- To address this need, we designed and tested a portable bat acoustic monitoring device.

Objectives

- To test the LoRa communication range of the device in hilly and cave environments.
- To evaluate the ultrasonic microphone's ability to capture bat calls in natural habitats.
- To verify data transmission from remote locations with no internet availability.
- To check reliable audio capture and system stability during field conditions.
- To validate the device for practical use in bat conservation research.

Related Sustainable Development Goal (SDG)

SDG 15 – Life on Land: Supports conservation of endangered bat species and protection of terrestrial ecosystems.



Societal Motivation and Need

Bats play a critical role in ecosystems, including:

- Natural pest control
- Pollination of plants
- Seed dispersal and forest regeneration

However, due to misconceptions and lack of public awareness, bat habitats are often disturbed or destroyed. The NGO identified the need for a non-intrusive monitoring system that could help researchers understand bat presence and behavior without harming the species.

Our involvement aimed to bridge this gap by assisting conservationists with a technological solution, while also understanding the ground realities of wildlife conservation

Field Activities

Site Visits and Observation

The field visits involved trekking through forested and hilly regions of Kolar, including Devarayanasangamudra Hill and Hanumanalli Hill, to reach natural bat habitats. These areas consisted of cave-like environments with minimal human interference. During the visits, we observed challenges such as difficult terrain, limited accessibility, and lack of basic infrastructure. We also interacted with researchers and volunteers from Bat Conservation India Trust to understand practical conservation needs in remote locations.

Field Learning Experience

The trekking experience highlighted the physical effort involved in conservation work and the constraints of working in ecologically sensitive zones. It emphasized the need for solutions that are portable, reliable, and capable of functioning in no-internet environments.

Technical Field Testing Activities

Device Testing in Field Conditions: During the field visits, we tested the developed device under real environmental conditions. The focus was on verifying long-range communication and basic audio capture capability in hilly and cave-like areas without internet connectivity. The testing confirmed that the device could operate reliably in remote locations, making it suitable for bat monitoring and conservation use.



Role of Technology in Conservation

Rather than focusing on complex technical implementation, the project emphasizes the practical importance of technology in wildlife conservation. The device enables detection of bat presence through acoustic monitoring without disturbing their natural behavior. By allowing long-term monitoring, it reduces the need for repeated human visits to sensitive habitats, thereby minimizing ecological disturbance. The system is designed to operate in remote areas and allows researchers to access collected data through a user interface, supporting informed and data-driven conservation decisions.

This approach aligns with the conservation goals of protecting endangered bat species while preserving their natural habitats.

Social and Educational Impact

From a societal and environmental perspective, the project supports conservation of endangered bat species and promotes eco-friendly, non-invasive monitoring methods. It strengthens collaboration between academic institutions and conservation organizations and contributes to increased awareness about the ecological importance of bats.

From a student learning perspective, the field visits provided valuable exposure beyond classroom learning. Working in real environmental conditions helped us develop teamwork, adaptability, and problem-solving skills. The experience also reinforced how engineering knowledge can be applied meaningfully to address societal and ecological challenges.

Outcomes and Key Takeaways

1. Field-Based Validation:

The field visits enabled successful validation of the device in real bat habitats, confirming its suitability for operation in hilly, cave-like, and no-internet environments.

2. Practical Exposure:

Hands-on field testing provided valuable practical exposure, allowing students to experience real-world challenges beyond controlled laboratory conditions.

3. Expert Interaction and Knowledge Exposure:

Interaction with the conservation expert from **Bat Conservation India Trust** provided insights into bat behavior, conservation challenges, and real-world monitoring requirements, enhancing our understanding from an applied conservation perspective.



4. Bridging Technology and Conservation:

The activity successfully connected engineering knowledge with wildlife conservation needs, demonstrating how technology can be applied responsibly to support environmental protection.

5. Student Development and Social Awareness:

The experience strengthened teamwork, adaptability, and environmental sensitivity among students, fostering a sense of social responsibility and motivation to engage in conservation-focused initiatives.

Concluding Remarks

The field visits to Kolar provided a valuable combination of practical testing and societal learning. Testing the device in real bat habitats helped us understand its suitability for remote, no-internet environments, while interaction with the expert from Bat Conservation India Trust gave us important insights into bat conservation and real-world monitoring needs. Overall, the experience reinforced the importance of applying engineering knowledge responsibly for environmental and societal benefit.

Student Participation & Eligibility

SL No.	NAME	USN	Activity Points
1.	Nihal Soni	1RV23CS152	40
2.	Nikhil Parashuram Bakale	1RV23CS154	40

Both students have successfully completed Three Field Visits associated with this project and are **eligible for 40 AICTE Activity Points**.



Pictures of Field Visits:

